Kinematic Restitution Balancing (KRB)
A Low-Cost Analytical Fix for Energy Gain in Velocity-Level Impulse Solvers

Joshua Sideris
Gearbox2D
https://gearbox2d.com
josh.sideris@gmail.com

Abstract

In discrete physics simulations using velocity-level impulse solvers (Sequential Impulse / Projected Gauss-Seidel), energy gain is a common numerical artifact. This paper introduces **Kinematic Restitution Balancing (KRB)**, a per-constraint correction for that leak in both unilateral constraints (collisions) and bilateral constraints (joints). By compensating for force-induced velocity drift (Component A) and auditing potential-energy shifts during position correction (Component B), KRB removes the two usual analytic energy sources inside a single-pass SI/PGS pipeline, without a second position pass. It does not claim a global energy invariant; evaluation and limitations sections state the coupled-constraint gap that follows from the per-constraint audit.

1. Introduction

Velocity-level sequential-impulse (SI) solvers [Catto 2005] are the standard real-time rigid-body approach: constraints are solved after explicit force integration with a fixed iteration count. KRB is a drop-in *preSolve* energy audit for that pipeline. It does not add solver passes, replace PGS, or switch to position-based dynamics.

Two analytic sources inject artificial mechanical energy during constraint resolution.

Force-integration drift. With symplectic Euler integration, velocity is updated before the constraint solve:

$$v_{\text{solver}} = v_t + a_{\text{ext}} \cdot \Delta t. \tag{1}$$

Restitution and joint bias see v_{solver} instead of the true relative velocity v_{impact}. For a perfect bounce ($e = 1$), the launch velocity gains $a_{\text{ext}} \cdot \Delta t$ extra speed per frame. For a joint at rest, the solver sees gravity-induced velocity and fights it every frame at the velocity level.

Potential-energy shift from position correction. Baumgarte stabilization [Baumgarte 1972] feeds constraint error back as a velocity bias $v_{\text{bias}} = \beta C/\Delta t$, displacing bodies by Δh to resolve penetration or joint drift. That displacement increases potential energy (mgh) without reducing kinetic energy. The added PE converts to KE on later frames, causing runaway energy growth.

Split impulse and equivalent decoupled position passes [Catto 2014] keep position-correcting impulses off the physical velocity used for restitution, which removes the *immediate* bounce-from-correction artifact. They do not tax the potential-energy work of the remaining displacement Δh. Nonlinear Gauss–Seidel (NGS) position solvers likewise omit that audit [Catto 2024]. Component B closes that gap inside a single-pass SI profile. KRB is not a dual-pass temporal solver, variational integrator, or PBD/XPBD stack.

KRB applies an analytic PE/KE balance to the *expected* correction displacement Δh at constraint setup, for contacts and distance joints. The audit is per-constraint; coupled graphs can still exhibit offsets when one constraint forces work on another (see evaluation and limitations).

2. Method

KRB performs two independent corrections during constraint `preSolve`.

2.1. Component A: Force velocity compensation

Store the velocity increment from external forces during integration ($v_{\text{force}} = a_{\text{ext}} \cdot \Delta t$ per body) and compute the true relative normal velocity:

$$v_{\text{impact}} = v_{\text{relative}} - \left(v_{\text{force},B} - v_{\text{force},A}\right). \tag{2}$$

Component A applies to all constraints whenever symplectic Euler integration is used, independent of the bias method.

2.2. Component B: Kinematic energy balancing

Adjust the launch or bias velocity to account for work done by external forces over the expected correction displacement Δh:

$$v_{\text{surf}} = \sqrt{\max\left(0,\ v_{\text{impact}}^2 + 2(\mathbf{a}_{\text{ext}} \cdot \mathbf{n})\Delta h\right)}, \tag{3}$$

$$v_{\text{final}} = e \cdot v_{\text{surf}} = e\sqrt{\max\left(0,\ v_{\text{impact}}^2 + 2(\mathbf{a}_{\text{ext}} \cdot \mathbf{n})\Delta h\right)}. \tag{4}$$

For ground collisions ($\mathbf{a}_{\text{ext}} \cdot \mathbf{n} < 0$), Component B taxes launch velocity to pay for increased PE; for ceiling collisions ($\mathbf{a}_{\text{ext}} \cdot \mathbf{n} > 0$), it credits velocity for work against external forces.

2.3. Effective displacement prediction

When velocity and position phases are decoupled, the kinematic bounce velocity v_{launch} partially resolves penetration d before the position solver runs:

$$d_{\text{eff}} = \max\big(0,\, d - (v_{\text{launch}} \cdot \Delta t)\big). \tag{5}$$

Many engines clamp per-iteration position correction to $\Delta h_{\max}$. For example, Gearbox2D uses `MAX_POSITION_CORRECTION`. The audit must match the work actually performed:

$$\Delta h = \min(d_{\text{eff}}, \Delta h_{\max}) \cdot \Gamma, \tag{6}$$

where $\Gamma = 1 - (1 - \beta)^n$ for n position iterations with Baumgarte factor β.

3. Constraint application

3.1. Contacts and speculative gaps

For physical overlap ($d > 0$), both Component A and Component B are required.

Speculative contacts are created before overlap ($d < 0$, a gap) to prevent tunneling. Because no position-correction work occurs yet, $\Delta h = 0$ and Component B is omitted; Component A remains critical so gravity-induced velocity is not treated as impact speed.

Apply restitution bias only when restitution is enabled and either the corrected normal velocity exceeds a small threshold, or a speculative gap is closing faster than the gap width allows. Resting contacts must not receive an $e = 1$ launch bias.

3.2. Distance joints and the zero-velocity paradox

Distance joints target zero relative velocity along the rod. Omitting Component B entirely leaks energy when Baumgarte stretch correction does work against gravity. Component B must be applied, but capped so a resting joint is not paralyzed.

Let $v_{\text{bias}} = \beta C / \Delta t$ be the ideal correction velocity for constraint error C. The audited correction is

$$v_{\text{bias\_actual}} = \sqrt{\max\big(0,\, v_{\text{bias}}^2 - 2(\mathbf{a}_{\text{ext}} \cdot \mathbf{n})(\beta C)\big)}. \tag{7}$$

If kinetic energy cannot pay the PE tax, the joint retains a microscopic Baumgarte sag—bounded audit rather than infinite stiffness at rest. This paper's setup recipe covers contacts and *distance* joints; other joint types are outside the listed implementation.

4. Implementation

KRB runs at contact and distance-joint `preSolve`, before the velocity iteration loop. Prerequisites: symplectic Euler with per-body `forceVelocity` ($a_{\text{ext}}\Delta t$), Baumgarte position correction with factor β, position iteration count n, and penetration slop.

4.1. Contact setup

```
// Component A: true impact normal velocity
forceVn = dot(bodyB.forceVelocity - bodyA.forceVelocity, n)
relativeVn = vn - forceVn

vBounce = -e * relativeVn
shouldBounce = enableRestitution &&
  (relativeVn < -RESTITUTION_THRESHOLD ||
   (depth < 0 && relativeVn < depth / dt))

if (shouldBounce) {
  if (depth > 0) {
    depthAfterVelocity = max(0, (depth - slop) - vBounce * dt)
    cumulativeFactor = 1 - pow(1 - beta, n)
    expectedDisplacement =
      min(depthAfterVelocity, MAX_POSITION_CORRECTION) *
      cumulativeFactor
  } else {
    expectedDisplacement = 0  // speculative:  Component A only
  }
  accVn = forceVn / dt
  workTerm = 2 * accVn * expectedDisplacement
  vSurfSq = relativeVn * relativeVn + workTerm
  vFinal = e * sqrt(max(0, vSurfSq))
  bias = (depth < 0) ? -max(vFinal, depth / dt) : -vFinal
} else if (depth < 0) {
  bias = -depth / dt
} else {
  bias = 0
}
```

Listing 1. Contact `preSolve` bias (Component A + B).

4.2. Distance-joint setup

```
float forceVn =
  (bodyB->forceVelocity - bodyA->forceVelocity).dot(normal);
float v_bias_ideal = beta * C / dt;
float expectedDisplacement = v_bias_ideal * dt;
float accVn = forceVn / dt;
```

```
float workTerm = 2.0f * accVn * expectedDisplacement;
float v_bias_sq = v_bias_ideal * v_bias_ideal;
float adjusted_v_bias_sq =
  max(0.0f, v_bias_sq - workTerm);
float v_bias_actual = sqrt(adjusted_v_bias_sq);
bias = (v_bias_ideal > 0 ? v_bias_actual : -v_bias_actual) - forceVn;
```

Listing 2. Distance-joint `preSolve` bias.

During the velocity solve, use the stored bias; Component A's $-\mathrm{forceVn}$ is folded into `bias` so `relative_vn` is not double-corrected each iteration.

5. Evaluation

Four datasets. Figure 1 is a compile-time ablation of the Gearbox2D sequential-impulse solver: default KRB on versus the same sources built with `-DGEARBOX_DISABLE_KRB`. Figure 2 places that KRB-on build next to unmodified Box2D-WASM, p2.js, and Matter.js. Dataset C, an 8 h KRB-on continuation of the cradle and a contact-only enclosure check, has no figure; traces are in `data/dataset-c/`. Dataset D is a native KRB on/off wall-time ablation on the same binary (no energy figure); traces are in `data/timing/`. We do not report a patched-Box2D result.

For Datasets A–C, every scene runs for 600 s of simulation time at $dt = 1/60$, $e = 1$, zero friction and damping, sleep disabled. Mechanical energy is $E = E_k + E_p$. Dataset A includes rotational KE; Dataset B is translational only (the website adapters do not expose ω). On the overlapping Gearbox traces the two agree to about 10^{-6} for the floor bounce. The reported ratio is E/E_0 relative to the known start-of-run energy.

Table 1. Dataset A ablation: E/E_0 at 600 s (compile-time KRB on/off).

Scene	KRB	E/E_0	Win. mean	Outcome
Floor bounce	on	0.999	1.000	bounded
Floor bounce	off	6.53	climbing	runaway
High-pressure circle	on	—	1.00	stayed in box
High-pressure circle	off	—	—	escaped, step 161
Newton's cradle	on	1.07	1.05 after $t = 300$	bounded offset
Newton's cradle	off	1.79	climbing	secular gain

Contact-only scenes match Section 1. With KRB, the floor bounce stays at $E/E_0 = 1.000 \pm 0.001$ in every 60 s window and never exceeds its start height. Without KRB the same drop gains about $0.55\,E_0$ per minute and finishes at $E/E_0 = 6.53$, with the apex 50 length units above the release. The high-pressure enclosure is the same leak at a higher impact rate: KRB window means stay at 1.00 for the full run; without KRB the body leaves the box after 2.7 s at $E/E_0 \approx 10$. Instantaneous 1 Hz samples of the on-curve swing between 0.67 and 1.33 because they alias the bounce; the windowed mean is the drift signal.

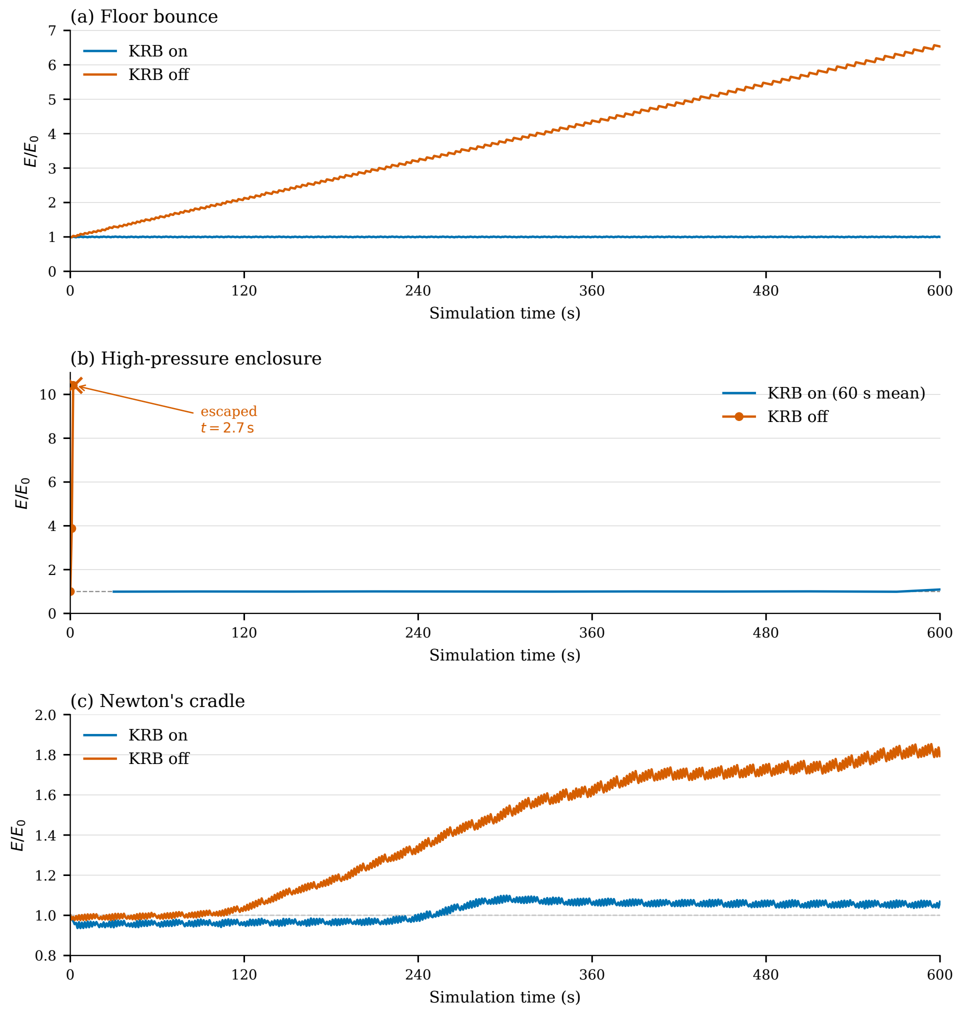

Figure 1. Mechanical energy ratio E/E_0 over $600\,\mathrm{s}$ of simulation time. (a) Single elastic floor contact. (b) High-rate enclosure; KRB on is a $60\,\mathrm{s}$ windowed mean, KRB off is raw $1\,\mathrm{Hz}$ samples terminating at tunneling ($t = 2.7\,\mathrm{s}$). (c) Newton's cradle (coupled contacts and joints). Vector original: `figures/fig-energy-ablation.pdf`.

The cradle is better with KRB but is not an invariant, which is the coupled-constraint gap in Section 6. Without KRB, E/E_0 climbs from 1.00 to 1.79 and outer-ball peak height from 1.35 to 2.43. With KRB, energy sits slightly low (~ 0.96) for four minutes, steps to ~ 1.07 around $t = 240$–$300\,\mathrm{s}$, then plateaus or slightly decays ($1.072 \to 1.054$). That is a bounded offset, not the linear SI ramp. A headless $8\,\mathrm{h}$ continuation (Dataset C) stays in that band: after the step, $E/E_0 \in [1.034, 1.093]$ with mean 1.053; the last hour matches the first hour after the step, and there are no further steps. The contact-only enclosure check in Dataset C stayed inside the box for $8\,\mathrm{h}$; $600\,\mathrm{s}$ windowed E/E_0 drifted $1.009 \to 0.977$ (slow loss, not the SI gain of Figure 1b).

Resting joint chains remain stationary with Component A enabled (no spurious force-drift correction at the velocity level). High-frequency spring modes are likewise not numerically damped by force-integration drift. Those two observations are qualitative and are not part of Figure 1.

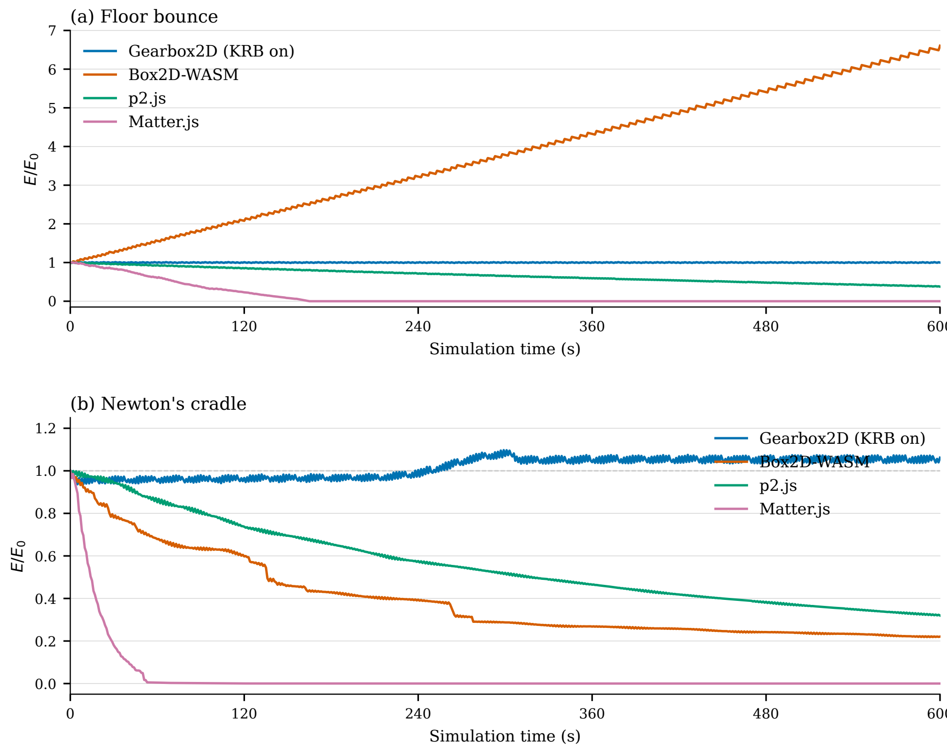

Figure 2. Mechanical energy ratio E/E_0 over $600\,\mathrm{s}$ on unmodified engines. (a) Elastic floor contact. (b) Newton's cradle. Gearbox2D is the default KRB-on WASM build; Box2D-WASM, p2.js, and Matter.js are unmodified whole-engine traces, not an in-engine KRB ablation. Vector original: `figures/fig-energy-engines.pdf`.

Dataset B is whole-engine behavior: iteration counts, split impulse, restitution thresholds, and default damping all differ. It is not an ablation of KRB inside those engines. Traces are in `data/dataset-b-*-600s.csv`.

The floor bounce is the contact leak from Section 1 in a production SI engine. Box2D-WASM climbs to $E/E_0 = 6.61$, matching Gearbox with KRB off (6.53) to a few percent. Gearbox with KRB on stays at 1.000 ± 0.001 in every $60\,\mathrm{s}$ window, as in Figure 1a. p2.js and Matter.js fail the other way: they dissipate, Matter reaching rest by $t \approx 160\,\mathrm{s}$.

The cradle is a different regime—elastic contacts plus joints—and the other engines lose energy rather than gain it. Matter is done by $t \approx 40\,\mathrm{s}$. Box2D-WASM falls in steps to 0.22 (restitution velocity threshold plus Baumgarte on the rods). p2.js drains smoothly to 0.32. Gearbox repeats Figure 1c: a $\sim 7\%$ step near four minutes, then a plateau. That is the empirical bound in Section 6, not a claim that the other engines share the SI gain of Figure 2a.

Energy loss on the cradle is not only a smaller amplitude. Unmodified Box2D, p2, and Matter lose the single-ball transfer: two balls leave together, or the pack splits, and the outer-

Table 2. Dataset B: E/E_0 at $600\,\mathrm{s}$ on unmodified engines.

Scene	Engine	E/E_0	Outcome
Floor bounce	Gearbox2D (KRB on)	0.999	bounded
Floor bounce	Box2D-WASM	6.61	runaway
Floor bounce	p2.js	0.376	slow loss
Floor bounce	Matter.js	≈ 0	dead by $t \approx 160\,\mathrm{s}$
Newton's cradle	Gearbox2D (KRB on)	1.062	bounded offset
Newton's cradle	Box2D-WASM	0.221	stepped loss
Newton's cradle	p2.js	0.320	slow loss
Newton's cradle	Matter.js	≈ 0	dead by $t \approx 40\,\mathrm{s}$

ball stroke drops. Gearbox keeps the one-at-a-time exchange for the full $600\,\mathrm{s}$. That is a qualitative observation; Figure 2b reports E/E_0.

A high-pressure enclosure ($g = 400$) was logged with the same harness and omitted from Figure 2. Matter rests on the floor within a second; p2.js does so within about $90\,\mathrm{s}$; Box2D-WASM locks to a two-value limit cycle (1.000 / 0.828 on alternate seconds) consistent with split impulse and a translation cap. That is a solver-survival test, not an energy comparison.

5.1. Cost (Dataset D)

The "Low-Cost" claim is checkable without a second solver pass. Per dynamic body at integrate, KRB stores $\mathrm{forceVelocity} = a_{\mathrm{ext}} \cdot \Delta t$ (two floats). Per contact `preSolve` when KRB is on: a $\mathrm{forceVn}$ dot, optional Γ, a `workTerm`, one `sqrt`, and a bias update. Per distance-joint `preSolve`: $\mathrm{forceVn}$, `workTerm`, one `sqrt`, with $-\mathrm{forceVn}$ folded into `bias`. There are zero extra PGS, velocity, or position iterations—not a second position pass or coupled-island PE tax. The paper recipe lists contacts and distance joints; whole-step timings below may include hinge Component A (engine-only).

Dataset D times `world.step()` on the same native `g++` binary with compile-time KRB on versus `-DGEARBOX_DISABLE_KRB` (`BENCH_FLAGS`: `-O3 -march=native -mfma -pthread -DGEARBOX_MT`). The product ships WASM; these numbers are a native ablation a reviewer recaptures via `make log-timing`. This is *not* a Gearbox-versus-Box2D CPU comparison (Dataset B is whole-engine energy, not in-engine ablation).

Scenes reuse Dataset A geometry for floor bounce, the high-pressure enclosure, and the cradle ($dt = 1/60$, $e = 1$, sleep off). Protocol: 100 warmup steps, then 1000 timed steps; three repeats; mean and range across repeats in ms/step (capture: i9-9900KF, g++ 13.3.0, `velocity_iterations`$= 50$, `position_iterations`$= 10$). We do not report a vanity percent speedup. Floor bounce and cradle on/off sit in the same scatter band; the enclosure shows a larger on/off spread ($\sim\ 0.012$ versus $\sim\ 0.006\,\mathrm{ms/step}$) but both remain $\ll 0.02\,\mathrm{ms/step}$. A denser `large-stack` row (not Dataset A) is included for contact load; on/off also overlap ($\sim 0.38\,\mathrm{ms/step}$).

Table 3. Dataset D: native KRB on/off wall-time per `world.step()` (mean and repeat range, ms/step). Compile-time ablation with `BENCH_FLAGS`; not Box2D. Traces: `data/timing/timing-summary-{krb,nokrb}.csv`.

Scene	KRB on	KRB off	Notes
Floor bounce	0.010 (0.005–0.018)	0.009 (0.005–0.017)	Dataset A
High-pressure circle	0.012 (0.012–0.013)	0.006 (0.006–0.007)	Dataset A
Newton's cradle	0.048 (0.047–0.048)	0.049 (0.047–0.050)	Dataset A
Large stack	0.377 (0.372–0.381)	0.375 (0.369–0.380)	not Dataset A

6. Limitations

KRB is a per-constraint audit, not a coupled one. When resolving one constraint (for example a horizontal collision) forces a second constraint to do work against gravity (for example a pendulum rod lifting the body), the isolated PE/KE tax may miss the systemic change in potential energy.

Newton's cradle is that coupled case. In Figure 1c and Figure 2b the KRB run shows a $\sim 7\%$ energy step near four minutes, then a plateau for the rest of the $600\,\mathrm{s}$ window—not a secular climb, and not a proof of a global invariant. Dataset C continues that KRB-on cradle to $8\,\mathrm{h}$ of simulation time ($dt = 1/60$, sampled every $10\,\mathrm{s}$; traces in `data/dataset-c/`). After the same four-minute step, E/E_0 stays in $[1.034, 1.093]$ (mean 1.053) through $t = 28800\,\mathrm{s}$; the last-hour mean is 1.053, identical to the hour after the step, and $60\,\mathrm{s}$ windowed means do not jump again. That is an empirical multi-hour bound for this scene, still not a global invariant, and still not shown for arbitrary contact/joint graphs.

The same $600\,\mathrm{s}$ of unmodified Box2D, p2, and Matter (Figure 2b) dissipate rather than gain; that is a different failure mode and does not close the coupled-audit gap. Variational integrators remain a different claim.

A coupled-constraint audit is future work. High-speed tunneling, long ill-conditioned chains, and other scenes that fail for reasons other than the two leaks in Section 1 are outside the claim. The no-KRB enclosure escape in Figure 1b is the SI leak at high impact rate, not a KRB failure.

7. Conclusion

Kinematic Restitution Balancing is a drop-in correction for velocity-level sequential-impulse solvers [Catto 2005]. Component A removes force-integration drift from restitution and joint bias; Component B taxes (or credits) the launch or bias velocity for the potential-energy work of the expected correction displacement Δh. Both apply to contacts and distance joints, at the cost of a few extra scalars per constraint setup (Table 3).

That is a narrower claim than a dual-pass solver or a reduced-coordinate formulation. Split impulse and NGS [Catto 2014, 2024] still hide Baumgarte bounce by keeping position work off the physical velocity; KRB does not replace that machinery, and it does not match their convergence behavior. It cancels the two analytic leaks in Section 1 inside a single-pass SI profile, with the coupled-constraint gap in Section 6 left open.

References

BAUMGARTE, J. Stabilization of constraints and integrals of motion in dynamical systems. *Computer Methods in Applied Mechanics and Engineering*, 1(1):1–16, 1972. URL: https://doi.org/10.1016/0045-7825(72)90018-7. 2

CATTO, E. Iterative dynamics with temporal coherence. Presented at the Game Developers Conference, San Francisco, CA, 2005. URL: https://box2d.org/files/ErinCatto_IterativeDynamics_GDC2005.pdf. 1, 9

CATTO, E. Understanding constraints. Presented at the Game Developers Conference, San Francisco, CA, 2014. URL: https://box2d.org/files/ErinCatto_UnderstandingConstraints_GDC2014.pdf. 2, 9

CATTO, E. Solver2d. Box2D blog post, February 2024. URL: https://box2d.org/posts/2024/02/solver2d/. 2, 9

Index of Supplemental Materials

Paths are relative to the study tree `studies/kinematic_restitution_balancing/`.

- `KRB_Whitepaper.md` — source of truth for claims and numbers.
- `jcgt/krb.tex`, `jcgt/krb.bib`, `jcgt/build-notes.md` — paper source, bibliography, TeX build notes.
- `figures/fig-energy-ablation.pdf` (and `.png`) — Figure 1 vector/raster.
- `figures/fig-energy-engines.pdf` (and `.png`) — Figure 2 vector/raster.
- `plot-energy-ablation.py`, `plot-energy-engines.py` — figure generators from committed CSVs.
- `data/README.md` — recapture notes and column layouts.
- `data/floor-bounce-krb.csv`, `data/floor-bounce-nokrb.csv`, `data/bounce-circle-krb.csv`, `data/bounce-circle-nokrb.csv` — Dataset A floor/circle traces.
- `data/cradle-krb.csv`, `data/cradle-nokrb.csv` — Dataset A cradle traces.
- `data/dataset-b-floor-bounce-600s.csv` — Dataset B floor bounce.
- `data/dataset-b-newtons-cradle-600s.csv` — Dataset B cradle.
- `data/dataset-c/cradle-krb.csv` — Dataset C 8 h cradle (no figure).
- `data/dataset-c/bounce-circle-krb.csv` — Dataset C 8 h enclosure (no figure).
- `data/timing/timing-summary-krb.csv`, `data/timing/timing-summary-nokrb.csv` — Dataset D native wall-time summaries.
- `log-energy.cpp` — headless logger for Datasets A and C.
- `log-timing.cpp` — Dataset D wall-time logger.

Author Contact Information

Joshua Sideris
Gearbox2D
https://gearbox2d.com
josh.sideris@gmail.com

Code and Data License

Supplemental source code and data named in the Index of Supplemental Materials are distributed under the ISC License (Open Source, BSD-class); see `LICENSE` in the Gearbox2D repository (https://github.com/JSideris/Gearbox2D).

Joshua Sideris, Kinematic Restitution Balancing: A Low-Cost Analytical Fix for Energy Gain in Velocity-Level Impulse Solvers, *Journal of Computer Graphics Techniques (JCGT)*, vol. ?, no. ?, 1–1, ????
????

Received:	August 21, 2026		
Recommended:	????-??-??	Corresponding Editor:	??? ???
Published:	????-??-??	Editor-in-Chief:	Alexander Wilkie